

Towards seamless integration of model, satellite and in-situ observation data with the VISION Toolkit Version 2

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Virtual Integration of Satellite and In-situ Observation Networks Toolkit v2

- Designed to enable efficient and accurate interfacing between **models (gridded and orderly)** and **observations (discrete, unstructured and often high in temporal resolution)**
- Supports satellite and *in-situ* observations (flight, ship, buoy or sonde etc. trajectories) to compare with any model data *as long as sufficiently CF Metadata Conventions compliant*
- Can facilitate systematic and consistent intercomparison, providing a practical approach to evaluation that reduces data volumes to help with the big data challenge of CMIP

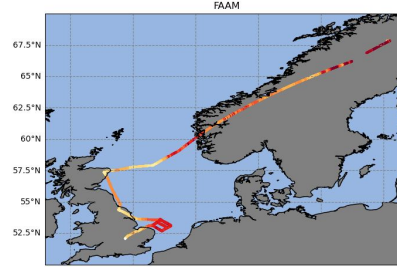
Codebase
(GitHub)



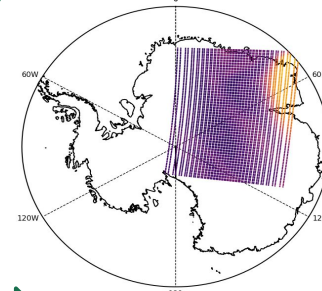
Documentation



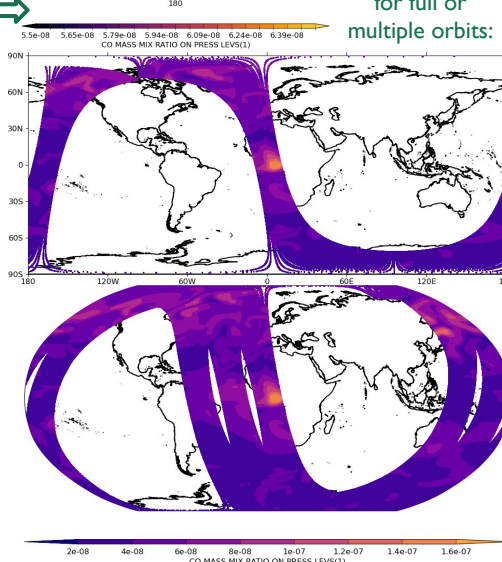
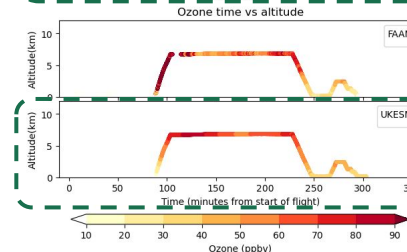
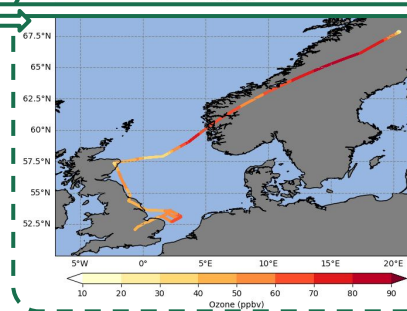
Flight example: FAAM aircraft path ↔ UKESM for model



Satellite example: swath or full orbit(s) ↔ Unified Model



Swath from a single dataset. Input multiple such datasets (e.g. full directory, or given glob pattern to select) at once for full or multiple orbits:



An open-source package for use on the command line or in Python, co-locating model data to observational geometry in a metadata-aware way, with much configurability:

